

LAXMI POORNAKALA MITTACOLA

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EDUCATION

Master of Computer Science, GPA - 3.30

December 2025

Illinois Institute of Technology, Chicago, IL

Bachelor of Technology in Computer Science, AI and Intelligent Process Automation Specialization, GPA - 3.8

April 2023

KL University, Vijayawada, AP, India

EXPERIENCE

Data Science Intern | Avishkar Tech Solutions

June - July 2021

- Developed machine learning models on structured datasets, improving prediction accuracy by ~30% through data preprocessing, feature engineering, and model evaluation.
- Built reproducible data pipelines for ingestion, cleaning, transformation, and validation to support ML experimentation and model development.
- Conducted statistical analysis and EDA to identify patterns, correlations, and data-quality issues and translate findings into business recommendations.
- Supported cloud-based ML deployment workflows, gaining exposure to model monitoring, automation, and iterative performance improvement.
- Created visualizations and reports to communicate model results, performance trade-offs, and technical findings to stakeholders.

PROJECTS

AI-Driven SDLC Automation | *AutoGen, LangGraph, LangChain, GenAI*

2025

- Designed and implemented a multi-agent AI system using LangChain, LangGraph, and AutoGen to automate software planning and documentation workflows from customer requirements.
- Built LLM-driven agent workflows integrating GPT-3.5, GPT-4, and GPT-4o-mini, with specialized agents responsible for development, testing, architecture, product, and documentation tasks.
- Designed and refined prompts and workflow logic, evaluating LLM outputs for consistency, completeness, and usefulness across different model configurations.

GitHub Issues Semantic Search | *Python, Elasticsearch, Embeddings, Docker*

2025

- Built an end-to-end semantic retrieval pipeline for GitHub Issues, including data ingestion, text preprocessing, embedding generation, Elasticsearch indexing, and vector-based retrieval.
- Implemented embedding-based semantic search with Elasticsearch to retrieve contextually relevant issues beyond traditional keyword matching.
- Containerized the retrieval workflow with Docker and evaluated representative semantic queries for retrieval relevance and consistency.

GitHub Repository Analytics & Forecasting Platform | *Python, TensorFlow/Keras, Prophet, Flask, React, GCP*

2025

- Developed LSTM, Prophet, and statistical forecasting models to predict repository activity trends across issues, commits, and releases.
- Compared deep learning and statistical approaches to evaluate forecasting performance and identify seasonal patterns and long-term activity trends.
- Integrated analytics and ML capabilities into a Flask/React application and deployed Dockerized services on Google Cloud Platform.

Mental Health Signals from Social Media | *Python, NLP, Machine Learning*

2025

- Developed an NLP and machine learning pipeline to analyze stress-related signals across 550K+ social media posts.
- Engineered sentiment, emotion, and text-based features and evaluated multiple classification models, including SVM, Random Forest, XGBoost, and Logistic Regression, achieving a best AUC of ~0.84.
- Evaluated model performance and generalization across datasets and analyzed results using statistical and visualization techniques.

SKILLS

- **Programming:** Python, Java, JavaScript, SQL, R, C
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Model Evaluation, NLP, Time-Series Forecasting, Deep Learning, LSTM
- **Generative AI & LLMs:** LLM Applications, Prompt Engineering, Agentic AI, Multi-Agent Systems, RAG, Embeddings, Semantic/Vector Search, LLM Evaluation, OpenAI, Ollama
- **AI/ML Frameworks:** TensorFlow/Keras, scikit-learn, XGBoost, LangChain, LangGraph, AutoGen, Pandas, NumPy, StatsModels, Prophet
- **Backend & Data Systems:** Flask, REST APIs, Django, GitHub API, Elasticsearch, MySQL, MongoDB, ETL/Data Pipelines
- **Cloud & DevOps:** GCP, Azure, AWS, Oracle Cloud, Docker, Git, GitHub, CI/CD Concepts, Cloud Deployment
- **Visualization & Analytics:** Matplotlib, Seaborn, Tableau, Statistical Analysis, Exploratory Data Analysis (EDA)

CERTIFICATIONS

- **Machine Learning with Python** - IBM, issued through Coursera
- **Deep Learning Specialization (v.2)** - DeepLearning.AI, issued through Coursera
- **Microsoft Certified:** AZ-400: Designing and Implementing Microsoft DevOps Solutions
- **Oracle Certified:** Oracle Cloud Infrastructure Certified Architect Associate
- **Aviatrix Certified:** Aviatrix Certified Engineer (ACE) Multicloud Network Associate (AWS, Azure, GCP, OCI)